Midhun Somanunnithan

Nürnberg, Germany · midhun.somanunnithan@fau.de · midhun9901.github.io/portfolio
linkedin.com/in/midhunsomanunnithan · github.com/midhun9901

Profile

M.Sc. Electromobility-ACES student at FAU (major: AI and autonomous driving) who combines deep learning for computer vision with hands-on manufacturing and materials-testing experience. Built a low-data image-retrieval pipeline (frozen DINOv2 + triplet-loss head: 32.7% → 78.3% mAP) and ran a 30-specimen experiment on automated speckle-pattern preparation for digital image correlation. Looking for a Master's thesis in computer vision or machine learning; based in Nürnberg, available immediately.

Research Experience

Research Project – Pattern Recognition Lab, FAU Erlangen-Nürnberg — Feb 2026 – present

Fine-grained retrieval of archaeological vase depictions · Supervisor: Mathias Zinnen · Code: github.com/midhun9901/vase-urn-project

- Built a PyTorch retrieval pipeline (feature extraction → metric-learning head → FAISS search → evaluation) and compared seven experiment families under one fixed vase-level split: ResNet-50 vs. DINOv2 features; triplet, ProxyAnchor and ArcFace losses; SAM crops; YOLO pose keypoints
- Best model: frozen DINOv2 ViT-S/14 with a small MLP head trained with triplet loss and hard-pair mining reached **78.3% mAP**, 77.4% Acc@1 and 100% Acc@10 on 53 held-out queries from unseen vases, up from 32.7% mAP with raw DINOv2 features, using only 200 training images
- Audited my own results: found that two SAM runs never applied their trained heads and that the training pool of a fine-tuned YOLO11 pose model overlapped the test set; fixed both pipelines and reran the clean baselines on all 253 images

Project Thesis – Chair of Manufacturing Technology (LFT), FAU Erlangen-Nürnberg — Apr 2026 – Sep 2026

Standardized DIC specimen preparation with automated pattern application · Supervisors: Prof. Marion Merklein, Adam Häckel

- Recommissioned an automated spray system for DIC speckle patterns: fixed unreliable paint-flow restarts with separate air valves; documented the Grbl G-code program and a cleaning routine
- Ran 10 spray configurations on 30 tensile specimens (DC04, DP600, DP1000), rated with ARAMIS pattern quality (PatQ), image-based measures and stress-strain curves; the recommended configuration (B2-C3) reached a mean PatQ of 7.26 on DP1000 vs. 4.19 for 18 existing hand-sprayed specimens
- Tested effects with Welch tests and Holm correction; the individual setting effects did not survive correction, and I reported this limit instead of claiming a significant improvement
- Designed a fixture (five iterations) that positions and masks twelve A50 tensile specimens in one row for batch spraying

Education

M.Sc. Electromobility-ACES – FAU Erlangen-Nürnberg — 2022 – present

Relevant courses: Pattern Recognition (2.0), Machine Learning for Engineers II: Advanced Methods (2.0), Machine Learning for Engineers I, Algorithms, Programming and Data Representation, Human-Computer Interaction

B.Tech. Mechanical Engineering – KTU Kerala, India — 2017 – 2021

Bachelor thesis: zone-aware automatic vehicle speed reduction with a Random Forest classifier, deployed with TinyML on a microcontroller

Selected Personal Projects

Deep Learning from Scratch — 2026

Reimplemented, following Karpathy's *Zero to Hero*, a scalar autograd engine and a six-layer character-level model with Kaiming initialization, BatchNorm and activation/gradient diagnostics · github.com/midhun9901/deep-learning-from-scratch

JARVIS: local AI voice assistant — 2025 – 2026

Always-on voice assistant: 5,000+ lines of Python (FastAPI, WebSockets), React UI, multi-LLM routing with provider fallback, background agents for memory consolidation · github.com/midhun9901/jarvis

Other Experience

Data Science & ML Intern – Entri Elevate — Oct 2023 – Dec 2023

Full data science lifecycle in Python: EDA, preprocessing, supervised models and evaluation; dashboards with Power BI and MySQL

Skills & Languages

Computer vision	metric learning (triplet, ProxyAnchor, ArcFace; hard-pair mining), DINOv2 / ViT, ResNet-50, SAM, YOLO pose fine-tuning, FAISS nearest-neighbour search, OpenCV
Evaluation	retrieval metrics (mAP, Acc@k), fixed class-level splits, leakage checks, hypothesis testing (Welch, Holm), DIC (ARAMIS)
ML & engineering	supervised learning, regression, ensemble methods, ANNs, TinyML; tensile testing, stress-strain and flow-curve analysis, fixture design, G-code
Programming	Python (PyTorch, NumPy, Pandas, scikit-learn), C, C++, MATLAB; Git/GitHub, SLURM on FAU's TinyGPU cluster, FastAPI, React
Languages	English (C1), German (B2), Malayalam (native)